

CN -- OSI Model (7 Layers)

[Target] Target: Name all 7 layers top-to-bottom + their KEY function + one protocol each in 40s.

Read time: 12 minutes

HIGH PRIORITY -- OSI is the #1 most asked CN topic in PSU interviews.

What Is OSI Model?

Open Systems Interconnection -- a conceptual framework that standardizes how different network systems communicate with each other in 7 layers.

Think of it as a postal system -- your letter (data) goes through multiple departments (layers), each adding/removing its own label (header/trailer), before reaching the destination.

The Mnemonic -- Never Forget This

Top to Bottom (Layer 7 - 1):

"All People Seem To Need Data Processing"

Application Presentation Session Transport Network Data Link Physical

Bottom to Top (Layer 1 - 7):

"Please Do Not Throw Sausage Pizza Away"

Physical Data Link Network Transport Session Presentation Application

Hinglish Version (Bottom to Top):

"Pehle Data Nikalte Hain, Transport Se Prapt Aata"

Physical, Data Link, Network, Transport, Session, Presentation, Application

Data Unit Names Per Layer

Application	}
Presentation	}
Session	}
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

Mnemonic: "Data Sent Neatly, Packets Find Bits" (top to bottom)

All 7 Layers -- Full Detail

Layer 7 -- Application Layer

"Tumhara interface hai duniya se" -- where YOU interact with the network

Details

Function

Network virtual terminal, Mail services, Directory services

What it does

Provides services to user applications; window to access network

Examples

Browsers, Skype, Messenger

Protocols

SMTP (email), HTTP/HTTPS (web), FTP (file), POP3 (email receive), SNMP, Telnet

Devices

-- (end-user software layer)

Layer 6 -- Presentation Layer

"Translator hai" -- converts data to a format the other side understands

Details

Also called
Translation Layer

Function
Translation (ASCII EBCDIC), Encryption/Decryption, Compression

What it does
Ensures data is in the right format for transmission and display

Protocols
MPEG (video), SSL/TLS (encryption), MIME (email format), XDR

Devices
--

Key 3 jobs: Translate + Encrypt + Compress

Layer 5 -- Session Layer

"Meeting arrange karta hai" -- manages the conversation

Details

Function
Session establishment, maintenance, termination; Synchronization; Dialog control

What it does
Sets up, manages, and tears down sessions between applications

Protocols
NetBIOS, SAP, PPTP, ADSP, PAP

Devices
Gateway, Servers

Key 3 jobs: Establish + Maintain + Terminate session

Layer 4 -- Transport Layer

"Delivery guarantee karta hai" -- end-to-end reliable delivery

Details

Data unit
Segment

Function

Segmentation & reassembly; End-to-end delivery; Service point addressing; Acknowledgement; Retransmission on error

What it does

Breaks large messages into segments, ensures complete delivery, reassembles at destination

Protocols

TCP (reliable), UDP (fast, unreliable), SPX

Devices

Firewall, Gateway

TCP vs UDP:

- TCP = reliable, connection-oriented, slower (handshake)
- UDP = unreliable, connectionless, faster (no handshake)

Layer 3 -- Network Layer

"GPS hai" -- finds the best path and handles IP addresses

Details

Data unit
Packet

Function

Routing; Logical Addressing (IP); Subnet traffic control

What it does

Places sender/receiver IP addresses in packet header; determines best route

Protocols

IPv4, IPv6, ICMP (ping), IPSEC, MPLS

Devices

Router, Brouter

Key point: Network layer adds IP addresses (logical addressing). Data Link layer adds MAC addresses (physical addressing).

Layer 2 -- Data Link Layer

"Dono paas ke stations ke beech" (node-to-node) -- local delivery, error-free

Details

Data unit
Frame

Function
Framing; Physical addressing (MAC); Error control; Flow control; Access control

What it does
Packages bits into frames; ensures error-free transfer between adjacent nodes

Protocols
PPP, ARP (IP MAC), Frame Relay, ATM, Fiber Cable

Devices
Switch, Bridge, Access Point

ARP = Address Resolution Protocol -- converts IP address to MAC address

Layer 1 -- Physical Layer

"Wires aur bits" -- actual physical transmission of raw bits

Details

Data unit
Bits (0s and 1s)

Function
Bit synchronization; Bit rate control; Physical topologies; Transmission mode

What it does
Converts signals to 0s and 1s; sends bits over the physical medium

Protocols
RJ-45, 100 Base Tx, ISDN

Devices

Hub, NIC (Network Interface Card), Cable, Modem, Wireless Repeaters

All 7 Layers -- Master Table

Layer
Name
Data Unit
Key Function
Protocols
Device

7
Application
Data
User interface to network
HTTP, FTP, SMTP, Telnet
--

6
Presentation
Data
Translate, Encrypt, Compress
SSL, MPEG, MIME
--

5
Session
Data
Establish/maintain/terminate sessions
NetBIOS, PPTP
Gateway

4
Transport
Segment
End-to-end delivery, segmentation
TCP, UDP
Firewall

3
Network
Packet
Routing + IP addressing
IPv4, IPv6, ICMP
Router

2
Data Link
Frame
Node-to-node, MAC, error control
ARP, PPP
Switch

1
Physical
Bits
Raw bit transmission
RJ-45, ISDN
Hub

OSI vs TCP/IP Model -- Quick Comparison

OSI (7 layers)
TCP/IP (4 layers)

Application (7)
Application

Presentation (6)
Application

Session (5)
Application

Transport (4)
Transport

Network (3)
Internet

Data Link (2)
Network Access

Physical (1)
Network Access

TCP/IP merges Application+Presentation+Session into one Application layer, and merges Data Link+Physical into Network Access.

Quick Cheat -- Layer Responsibilities in One Word

Layer 7 Application	USER
Layer 6 Presentation	TRANSLATE
Layer 5 Session	MANAGE
Layer 4 Transport	DELIVER (TCP/UDP)
Layer 3 Network	ROUTE (IP + Router)
Layer 2 Data Link	FRAME (MAC + Switch)
Layer 1 Physical	BITS (Hub + Wire)

Your 40-Second Script -- OSI Model

"OSI model has 7 layers. Bottom to top: Physical transmits raw bits, Data Link handles node-to-node delivery using MAC addresses and frames, Network handles routing and IP addressing using packets, Transport ensures end-to-end delivery using TCP or UDP in segments, Session manages connections, Presentation handles translation and encryption, Application is the user-facing layer with protocols like HTTP and FTP.
Mnemonic bottom-up: Please Do Not Throw Sausage Pizza Away."

Follow-Up Questions

Q: What is the difference between TCP and UDP?

TCP is connection-oriented, reliable, provides acknowledgement and retransmission -- used for HTTP, email, file transfer. UDP is connectionless, faster, no guarantee of delivery -- used for video streaming, DNS, gaming where speed & reliability.

Q: What is the difference between Switch and Router?

Switch operates at Layer 2 (Data Link) -- uses MAC addresses, connects devices within the same network. Router

operates at Layer 3 (Network) -- uses IP addresses, connects different networks.

Q: What is ARP?

Address Resolution Protocol (Layer 2) -- resolves an IP address to its corresponding MAC address. When a device knows the IP but needs the MAC to send a frame on the local network, it broadcasts an ARP request.

Q: Which layer adds IP address? Which adds MAC address?

Network Layer (3) adds IP address (logical). Data Link Layer (2) adds MAC address (physical).